

To: Human Resources Department
From: Mikhail Jacques
Subject: Summary of Professional Contributions and Activities

Dear HR Team,

I am writing to provide a comprehensive summary of my professional contributions and responsibilities at Israel Aerospace Industries (IAI) in support of my candidacy for promotion.

Since joining IAI in December 2015 as a Software Engineer, I have consistently delivered high-impact solutions across multiple departments, demonstrating technical depth and leadership. I have led and participated in full software development life cycles—from initial requirements and architectural design through implementation, integration, and documentation—delivering reliable, real-time, and research-grade applications.

UAV Real-Time Software Department (Dec 2015 – Oct 2020)

During this period, I spearheaded the development of new non-avionic control logic in C, collaborating closely with avionics engineers to co-author specifications and technical documentation. My work primarily focused on governing critical UAV subsystems, including engine, fuel, and fuel pump operations.

I also took the initiative to refactor and elevate large portions of the UAV plane-control codebase, which had accumulated technical debt and architectural drift. Among the key contributions:

- Rebuilt and standardized entire modules, including Digital I/O, Serial Lines, Non-Volatile Memory, and AVC-to-AVC communication.
- Consolidated scattered logic and files into a clean, hierarchical structure that now follows high-cohesion, low-coupling principles.
- Introduced code efficiency improvements optimized for PowerPC CPUs.
- Re-architected the watchdog logic and implemented robust exception logging.
- Improved code readability and maintainability through consistent naming conventions, modularization, and rigorous documentation.
- Removed obsolete, redundant, and dead code throughout the codebase, significantly improving maintainability.
- Upgraded development environment compatibility to GHS Multi 6.1.6 and 7.1.4.
- Developed a new Avionics Monitoring (Avim) application in C++ using the Qt framework.

Algorithmics Department (Nov 2020 – Dec 2021)

I designed and implemented, from the ground up, a central management module in C++ for a Windows-based Image Landing research project. The work included requirements analysis, software architecture, implementation, testing, and documentation.

Robotics Department – Elta Systems (Jan 2022 – July 2022)

I designed and implemented another C++ Windows-based central module for the World Perception project. My work integrated and managed communication among various robotic subsystems in a research environment.

Integration Department (July 2022 – July 2023)

I explored cutting-edge technologies and delivered several experimental applications in C, C++, C#, and Python, supporting a range of research initiatives.

Engineering Department (July 2023 – Jan 2024)

I designed, developed, and deployed a real-time, Windows-based Ground Video-Data Multiplexer (GVDM) message-routing system, now in active use by the Israeli Air Force. This high-performance system was built from scratch in C++ and involved full-cycle development from requirements through deployment.

Current Role – Integration Department (Jan 2024 – Present)

I am currently developing a Linux-based C++ signal-processing application, continuing to contribute to mission-critical research and development initiatives.

Mentorship and Collaboration

Throughout my tenure, I have consistently collaborated with avionics engineers, provided technical mentorship to junior developers, and facilitated cross-team knowledge transfer. My contributions extend beyond coding to include fostering clarity in documentation, code comprehension, and architectural reasoning.

I am proud of my continuous commitment to engineering excellence, innovation, and mentorship at IAI. I believe that my diverse technical achievements, initiative, and leadership qualify me for a role of greater responsibility, and I respectfully submit this record in consideration for promotion.

Thank you for your time and attention.

Sincerely,

Mikhail Jacques

Software Engineer, IAI

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I have worked as a software engineer since July 2012. I have been employed as a software engineer at IAI since December 2015.

From December 2015 until now, I have worked as a software engineer in the UAV plane-control real-time software department. During this time, I developed and implemented a new non-avionic programming logic from scratch in the C programming language. I co-authored requirements and related technical documentation with avionics engineers. The non-avionic programming logic, for the most part, governed and monitored the engine, fuel, and fuel pump functionalities. On top of that, I took the initiative to improve what was within my power in the UAV plane-control real-time software, which was a real mess, to say the least. The following is a list of things I am responsible for fixing and upgrading. The list is a highlights-only compilation and is not exhaustive.

- Rebuilt, standardized the entire Avim module,
- Rebuilt, standardized and documented the entire digital inputs/outputs module,
- Rebuilt, standardized and documented the entire serial lines module,
- Rebuilt and standardized the entire AVC-to-AVC communication module,
- Rebuilt, standardized and documented the entire Non-Volatile memory (NvRAM) module,
- Developed, integrated and tested new programming logic that now saves both the first and the last exception information in NvRAM,
- Improved and standardized Watchdog programming logic,
- Added antenna (it has neither been tested nor formally documented yet),
- Rebuilt and standardized the entire directional antenna module (it has neither been tested nor formally documented yet),
- Standardized tasks and connections,
- Standardized code writing signature where applicable and possible,
- Standardized the code as much as I am allowed by ICD and flight control constraints,
- Consulted with Elta Systems Ltd. documentation and with local hardware engineers regarding proper terminology and then re-implemented it to adhere to commonly accepted standards,
- Removed obsolete OPT code,
- Removed tons of dead code,
- Removed unused files,
- Removed duplicate files,
- Removed redundant tasks,
- Removed duplicate definitions within the files,
- Re-aligned code to look the same throughout the entire project,
- Renamed and standardized files to reflect the essence of their purpose,
- Renamed and standardized variable names,
- Optimized references to files and consolidated them in one place,
- Consolidated data in files with strict adherence to key software engineering principle of high coherence and low coupling,
- Improved code efficiency from the PowerPC CPU perspective,

- Added new features and improved operational program safety,
- Upgraded code to develop and compile using GHS Multi 6.1.6. and Multi 7.1.4 as opposed to old Multi 5.0.6,
- Reconstructed the entire project's hierarchical tree structure because it was obsolete, messy and inefficient,
- Relocated all address spaces into one folder, namely APPLS
- Relocated all the software modules that deal with devices (LRUs) from wherever they were scattered into the Devices folder under the Main address space,
- Relocated all the ICD-generated files from wherever they were scattered into the dedicated ICD directory,
- Discovered numerous code defects along the way and fixed them,
- Documented code via self-descriptive names of files, functions and variables,
- Developed and partially documented a code standard that anyone working on MK2 software or its derivatives should adhere to.

In addition, in the fall of 2017, I designed and developed a new Avionics monitoring (Avim) application from scratch in C++ using the Qt framework.

From November 2020 until December 2021, I worked in the Algorithmics department. During that time, I gathered and analyzed requirements, designed and developed a central management module from scratch in C++, and tested and documented it for the Windows-based Image Landing research project.

From January 2022 to July 2022, I worked in the Robotics department at Elta Systems. During that period, I analyzed requirements, designed and developed a Windows-based central management and communication module from scratch in C++, and tested and documented it for the World Perception research project.

From July 2022 until July 2023, I primarily researched new technologies and implemented a few small research applications for the Integration department, using a wide variety of technologies and programming languages, including C, C++, C#, and Python.

From July 2023 until January 2024, I gathered and analyzed requirements, prepared requirements documentation, designed the architecture, developed a real-time Windows-based production application for the Engineering Department from scratch in C++, tested, integrated, and documented a Ground Video-Data Multiplexer (GVDM) message-routing application that is currently used by the Israeli Air Force.

Over the years, I also assisted various avionics engineers with documentation and code understanding, and I tutored and mentored less-experienced software engineers across various technologies and coding assignments.

Right now, I am developing a Linux-based signal processing application in C++ for the Integration Department.